CS5486 Intelligent Systems

Introduction to Intelligent Systems & The Development History of Artificial Intelligence

Fangyu Zhang

Department of Computer Science

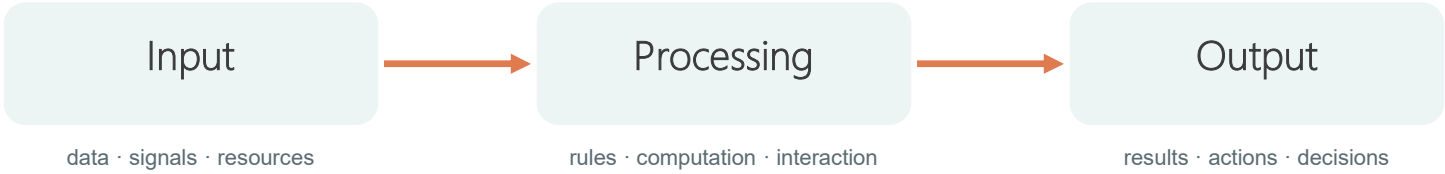
City University of Hong Kong

Defination of system

A system is a set of interrelated components that interact with each other — and with their environment — to perform a function or achieve a goal.

— A general working definition

Inside a system



Systems are everywhere — a car, a computer, the human body.

Defination of Intelligent system

Intelligent systems

> "A system is an intelligent system if it exhibits some intelligent behaviors."
>
> — Definition used in this course

Examples in this course

Expert systems · Fuzzy systems · Neural networks

Defination & Classification of AI

Definition: Artificial Intelligence (AI) is the simulation of human intelligence by machines.

AI can be categorized by capability level:

Narrow AI (Weak AI)

- Specialized in one or multiple tasks
- Most AI today (e.g., siri, Google Translate)

General AI (AGI)

- Human-level intelligence across domains
- AGI requires a profound comprehension of the real physical world.
- Still theoretical

Superintelligent AI

- Exceeds human intelligence
- Exists only in theory or fiction

AI vs. Intelligent Systems

Artificial Intelligence

The field of study — theories, methods, and algorithms that enable machines to simulate intelligent behavior.
A toolbox of techniques.

builds

Intelligent Systems

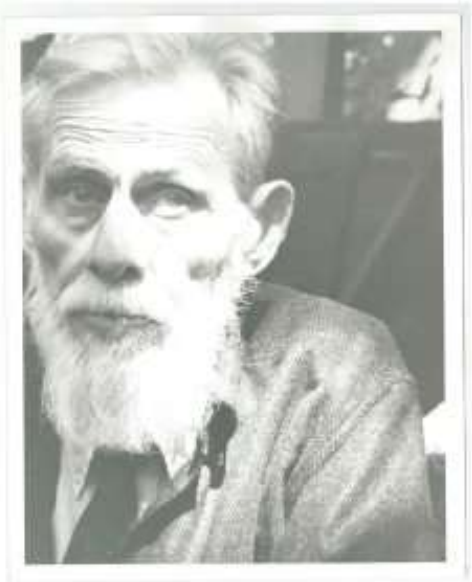
Working systems built using AI techniques — software or hardware that performs intelligent tasks in practice.
The product from the toolbox.

How did AI techniques evolve over time?
→ **The Development History of Artificial Intelligence**

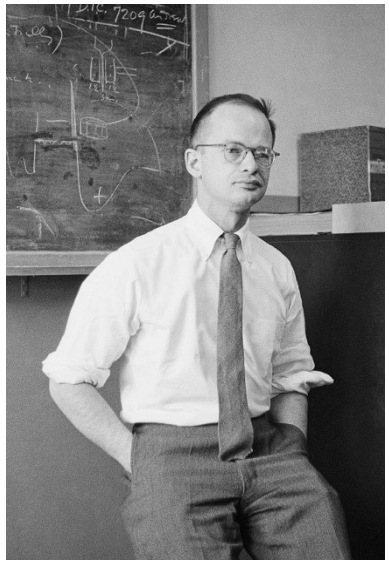
豆包AI生成

The Origins of AI (1940s–1950s)

1943: Warren Sturgis McCulloch and Walter Pitts publish "A Logical Calculus of the Ideas Immanent in Nervous Activity", the first mathematical description of an artificial neural networks.



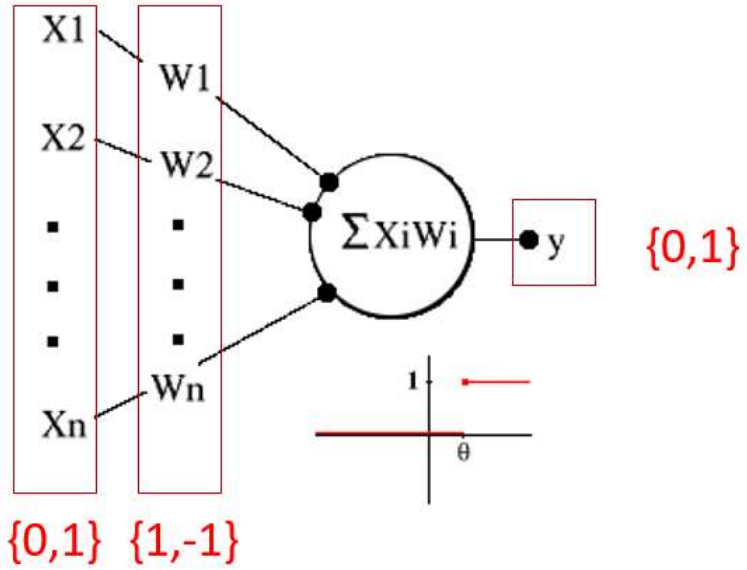
Warren Sturgis McCulloch (1898-1969)



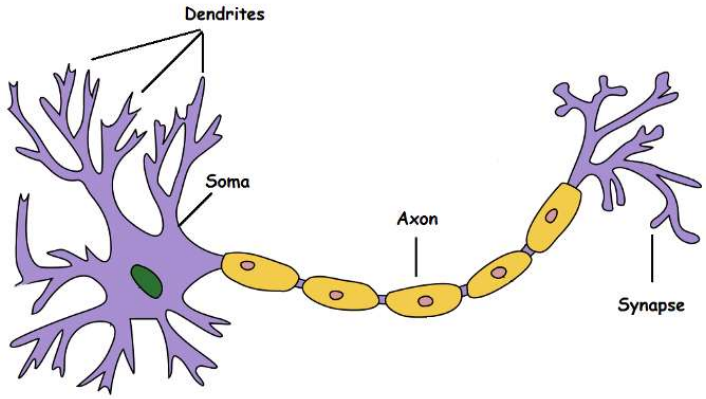
WALTER PITTS (1923-1969)

The Origins of AI (1940s–1950s)

McCulloch-Pitts Neuron



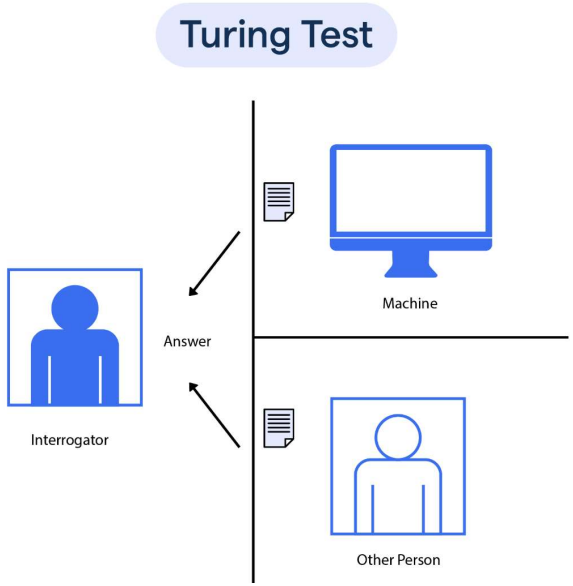
Biological Neuron



The Origins of AI (1940s–1950s)

1950: Alan Turing published "Computing Machinery and Intelligence", which proposes the Turing test as a measure of machine intelligence.

- The interrogator's job is to determine which is the human and which is the machine by asking any kind of questions.
- If the machine can fool the interrogator into thinking it's human as often as (or more than) the actual human, it is said to pass the Turing Test.



The Origins of AI (1940s–1950s)

1956: Dartmouth Conference held in New Hampshire, USA

The term Artificial Intelligence (AI) was coined by John McCarthy for this conference.



Golden Era I (1956–1974)

In 1958, Rosenblatt proposed the “Perceptron”, a machine learning algorithm used to determine whether an input belongs to one class or another



Psychological Review
Vol. 65, No. 6, 1958

THE PERCEPTRON: A PROBABILISTIC MODEL FOR INFORMATION STORAGE AND ORGANIZATION IN THE BRAIN[1]

F. ROSENBLATT

Cornell Aeronautical Laboratory

If we are eventually to understand the capability of higher organisms for perceptual recognition, generalization, recall, and thinking, we must first have answers to three fundamental questions:

1. How is information about the physical world sensed, or detected, by the biological system?
2. In what form is information stored, or remembered?
3. How does information contained in storage, or in memory, influence recognition and behavior?

and the stored pattern. According to this hypothesis, if one understood the code or "wiring diagram" of the nervous system, one should, in principle, be able to discover exactly what an organism remembers by reconstructing the original sensory patterns from the "memory traces" which they have left, much as we might develop a photographic negative, or translate the pattern of electrical charges in the "memory" of a digital computer. This hypothesis is appealing in its simplicity and ready intelligibility, and a large family of theoretical brain

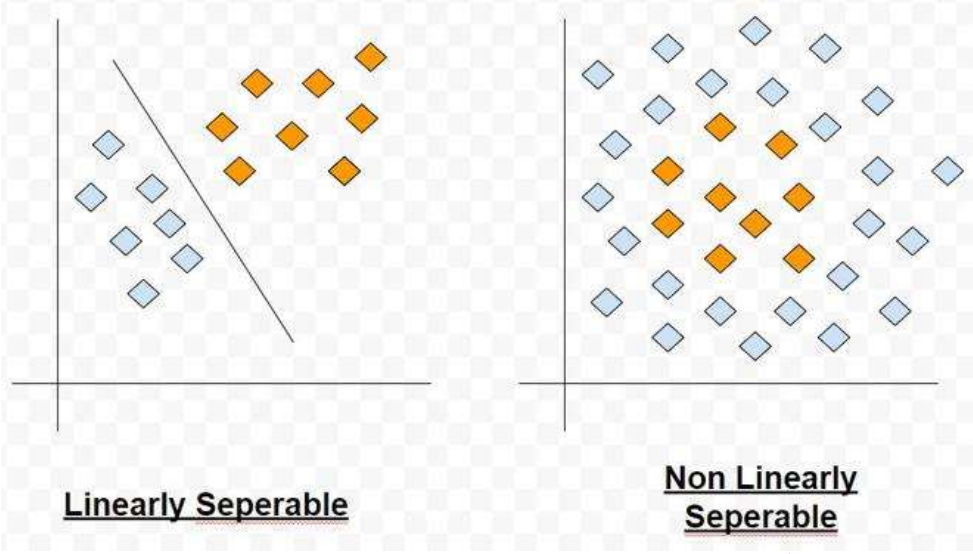
Paper review - (1958)

Rosenblatt, Frank. "The perceptron: a probabilistic model for information storage and organization in the brain."

Golden Era I (1956–1974)

A single perceptron can learn to classify any half-space.

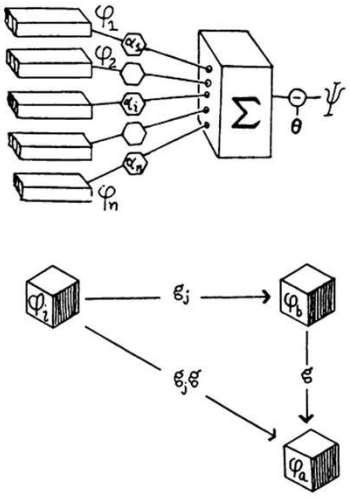
It cannot deal with linearly nonseparable vectors, such as XOR problem.

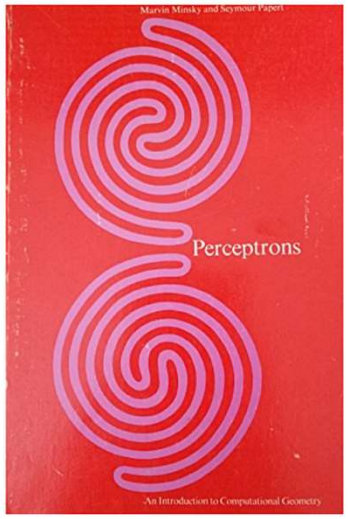


Multi-layer perceptrons are needed

Golden Era I (1956–1974)

In 1969, a famous book entitled Perceptrons by Marvin Minsky and Seymour Papert showed that it was impossible for these classes of network to learn an XOR function. The often-miscited Minsky and Papert text caused a significant decline in interest and funding of neural network research







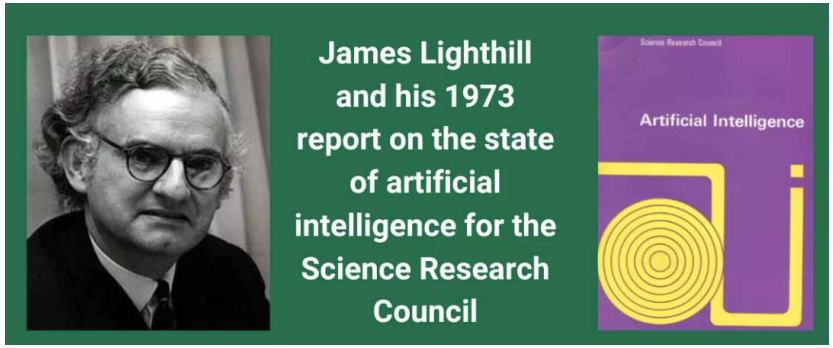
M. Minsky



S. Papert

First AI Winter (1974–1980)

1973: "Artificial Intelligence: A General Survey" by James Lighthill (commonly known as the Lighthill Report) directly contributed to the first AI winter



Key Conclusions:

- AI had failed to deliver on its promises
- Progress was limited to "toy problems" in narrow academic settings
- AI systems lacked scalability and general intelligence
- The field was fragmented and lacked theoretical rigor

Result: Massive Funding Cuts

First AI Winter (1974–1980)

Causes:

- Funding cuts (e.g., Lighthill report in the UK)
- Overhyped expectations
- Hardware limitations

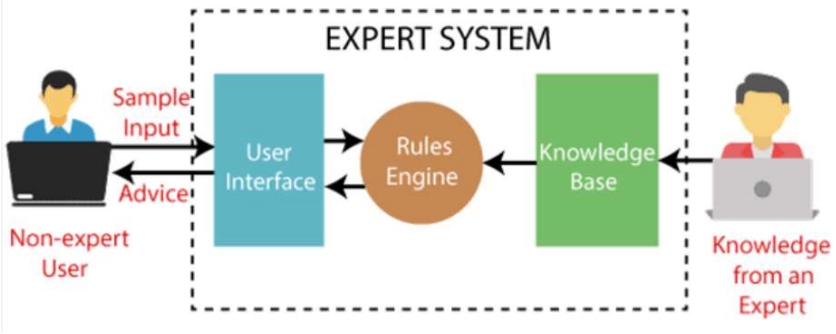
AI research lost credibility and support.

Expert Systems & Golden Era II (1980s)

The Second AI Boom emerged in the early 1980s and was driven by several key factors:

- Rise of Expert Systems

  Expert systems simulated human reasoning using a knowledge base and inference engine.



They were applied in medicine, finance, geology, etc.:

- MYCIN: Diagnosed bacterial infections and recommended antibiotics.
- XCON: Helped Digital Equipment Corporation (DEC) configure computer systems efficient

Expert Systems & Golden Era II (1980s)

The Second AI Boom emerged in the early 1980s and was driven by several key factors:

- Commercial Success

  Major companies (e.g., IBM, DEC, Xerox) began investing heavily in AI.

  AI tools promised to improve productivity and reduce costs.

- Government & International Initiatives

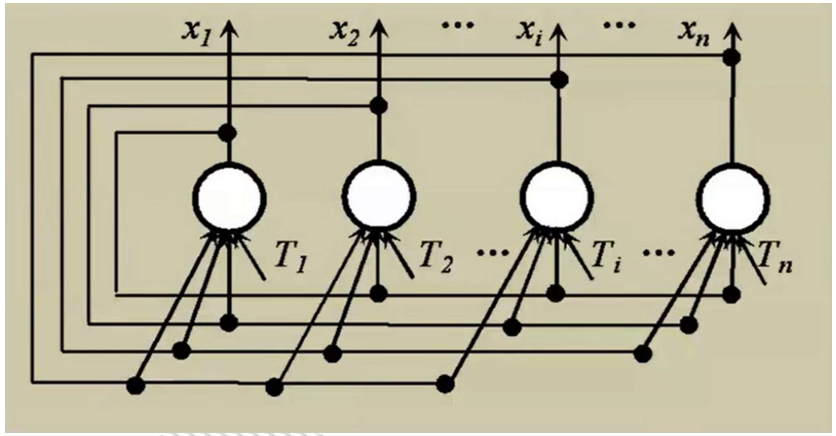
  In 1982, Japan launched the Fifth Generation Computer Systems Project, investing hundreds of millions to develop AI with logic programming (e.g., Prolog) and parallel computing.

  This triggered a "technological race", prompting U.S. and European governments to increase funding for AI research.

Expert Systems & Golden Era II (1980s)

After the "AI winter" following the limitations of the Perceptron (1969), research on neural networks had slowed.

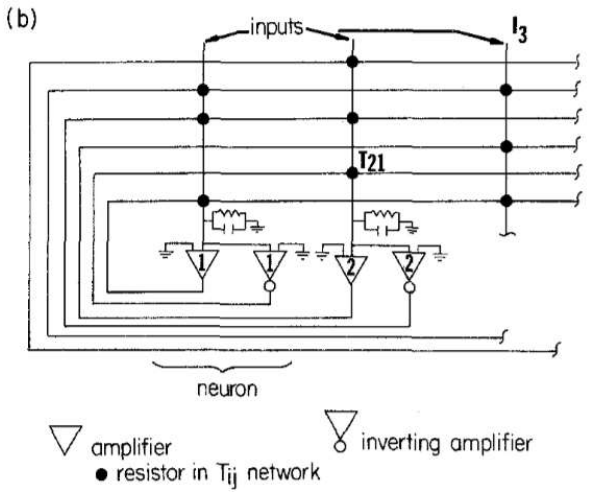
In 1982, Hopfield Network was proposed, reigniting academic and practical interest in neural computation. Hopfield Network is a kind of recurrent neural network capable of associative memory, pattern completion, and optimization.



Expert Systems & Golden Era II (1980s)

Applications of Hopfield Network include:

- Solving optimization problems (e.g. Traveling Salesman Problem (TSP) problem)



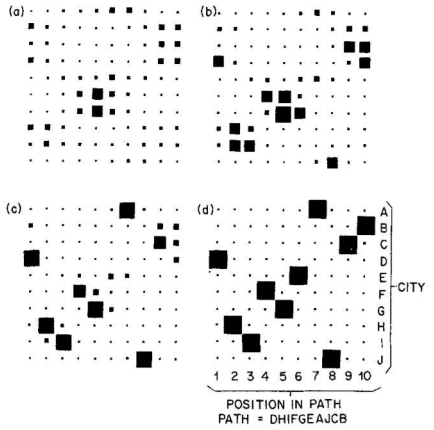


Fig. 2a–d. The convergence of the 10-city analog circuit to a tour. The linear dimension of each square is proportional to the value of V_{Xi}. **a**, **b**, **c** intermediate times, **d** the final state. The indices in **d** illustrate how the final state is decoded into a tour (solution of TSP)

Expert Systems & Golden Era II (1980s)

Applications of Hopfield Network include:

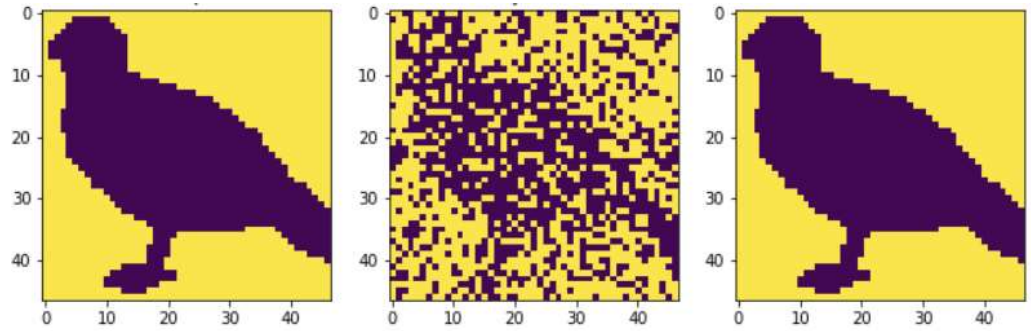
- Image Denoising and Restoration

Application Scenario

Hopfield networks can be used to restore corrupted or noisy images by leveraging stored ideal versions of images.

Example

If an image has missing pixels or noise, a Hopfield network trained on the original image can reconstruct the missing parts by converging to the closest stored pattern. This method is useful in **medical imaging, satellite image reconstruction, and old photograph restoration**.



Expert Systems & Golden Era II (1980s)

Applications of Hopfield Network include:

- Speech Recognition

Application Scenario

Hopfield networks can be used in **speech pattern recognition**, filtering noisy speech signals and recognizing key phonemes.

Example

If a stored pattern represents a clean speech signal, a noisy input can be corrected to match the stored version, improving clarity. This is useful in **voice assistants, hearing aids, and automatic transcription services**.

- Content Addressable Memory

Application Scenario

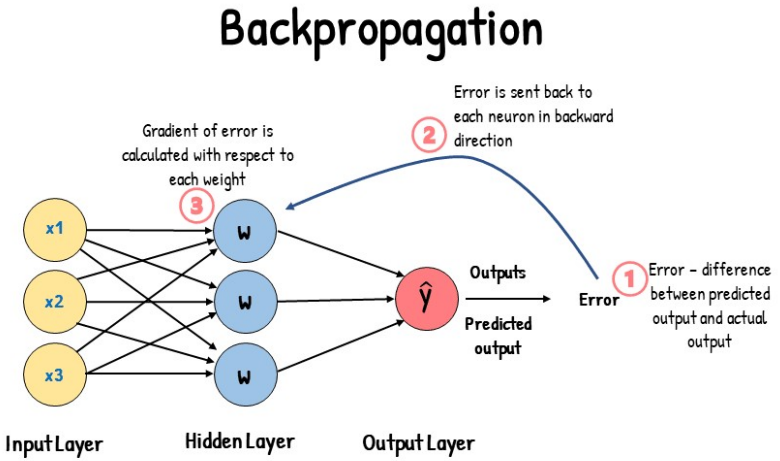
Hopfield networks work as a special type of memory that retrieves complete information based on partial input, similar to human memory recall.

Example

After storing multiple patterns, when given incomplete information, the network uses an energy minimization process to retrieve the most similar stored pattern. This is useful in applications like database searches and information retrieval.

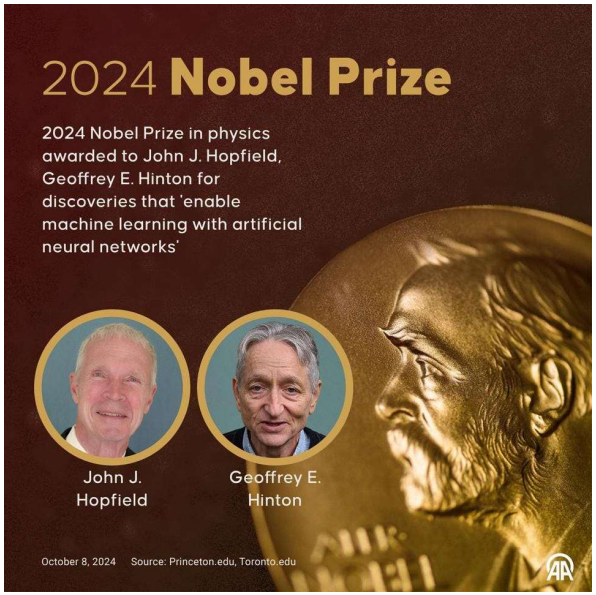
Expert Systems & Golden Era II (1980s)

Then, in 1986, Geoffrey Hinton and colleagues proposed the backpropagation algorithm, enabling efficient training of multi-layer neural networks and laying the foundation for modern deep learning. This breakthrough is the core of today's deep learning revolution.



Expert Systems & Golden Era II (1980s)

Geoffrey Hinton and John Hopfield were jointly awarded the 2024 Nobel Prize in Physics for their foundational discoveries and inventions in the field of artificial neural networks and machine learning.





Second AI Winter (1987–1993)

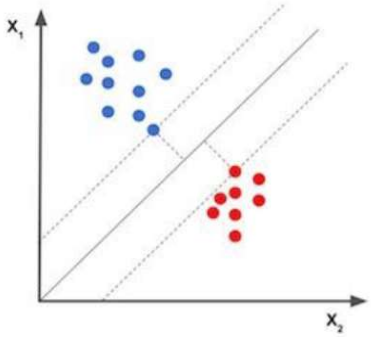
Causes:

- Limitations of Expert Systems
  - Brittle: Could not handle unexpected situations.
  - Knowledge acquisition bottleneck: Extracting expert knowledge was difficult and time-consuming.
  - Maintenance nightmare: Updating rule-based systems became unmanageable as they grew.
- Hardware Limitations
  - Computers at the time lacked the power to run large, complex AI systems effectively.
- Market Collapse
  - Many AI startups and hardware vendors (like Lisp machine makers) went bankrupt.
  - Investors pulled out, and AI research funding dried up.
- Failure of Japan's Fifth Generation Project
  - International confidence in AI was shaken.

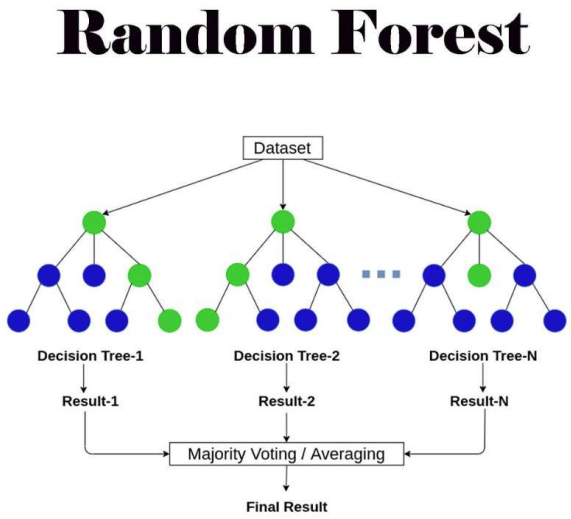
Rise of Machine learning(1990s–2010)

Transfer to statistical learning

Key algorithms: SVMs, decision trees, random forests, etc.



SUPPORT VECTOR MACHINE



Rise of Machine Learning (1990s–2010)

1997: IBM's Deep blue defeats chess world champion Garry Kasparov.

Based on minimax search algorithm and Alpha-Beta pruning, Deep Blue could evaluate up to 200 million positions per second, and search 10–20 moves deep.



Rise of Machine Learning (1990s–2010)

2006: Geoffrey Hinton proposes Deep Belief Networks, the Starting Point of Deep Learning

LETTER Communicated by Yann Le Cun

A Fast Learning Algorithm for Deep Belief Nets

Geoffrey E. Hinton
hinton@cs.toronto.edu
Simon Osindero
osindero@cs.toronto.edu
Department of Computer Science, University of Toronto, Toronto, Canada M5S 3G4

Yee-Whye Teh
tehyw@comp.nus.edu.sg
Department of Computer Science, National University of Singapore, Singapore 117543

We show how to use "complementary priors" to eliminate the explaining-away effects that make inference difficult in densely connected belief nets that have many hidden layers. Using complementary priors, we derive a fast, greedy algorithm that can learn deep, directed belief networks one layer at a time, provided the top two layers form an undirected associative memory. The fast, greedy algorithm is used to initialize a slower learning procedure that fine-tunes the weights using a contrastive version of the wake-sleep algorithm. After fine-tuning, a network with three hidden layers forms a very good generative model of the joint distribution of handwritten digit images and their labels. This generative model gives better digit classification than the best discriminative learning algorithms. The low-dimensional manifolds on which the digits lie are modeled by long ravines in the free-energy landscape of the top-level associative memory, and it is easy to explore these ravines by using the directed connections to display what the associative memory has in mind.

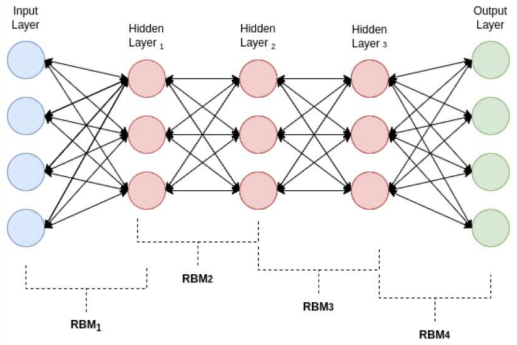
1 Introduction

Learning is difficult in densely connected, directed belief nets that have many hidden layers because it is difficult to infer the conditional distribution of the hidden activities when given a data vector. Variational methods use simple approximations to the true conditional distribution, but the approximations may be poor, especially at the deepest hidden layer, where the prior assumes independence. Also, variational learning still requires all of the parameters to be learned together and this makes the learning time scale poorly as the number of parameters increases.

We describe a model in which the top two hidden layers form an undirected associative memory (see Figure 1) and the remaining hidden layers

Neural Computation **18, 1527–1554** (2006) © 2006 Massachusetts Institute of Technology





Deep Learning Boom (2012–2018)

2012: AlexNet (by Hinton's team) achieved a breakthrough victory in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC). AlexNet reduced image classification error by ~10%, igniting the deep learning boom.



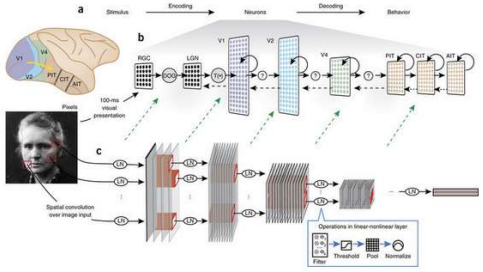
Deep Learning Boom (2012–2018)

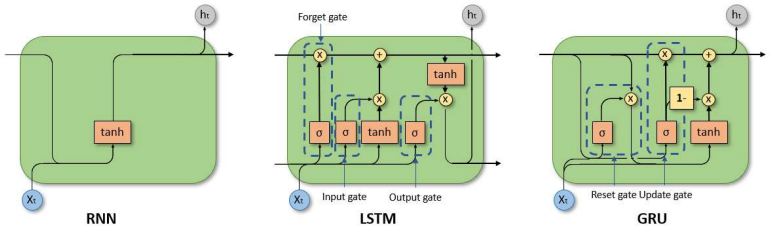
Key Technologies:

- Convolutional Neural Networks (CNNs) for computer vision tasks (e.g. AlexNet, Visual Geometry Group Network (VGG), Residual Network (ResNet))

- Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for sequential data modeling (e.g. speech, text, time series)

- Graphics Processing Units (GPUs) significantly boosted training speed and enabled large-scale neural network experiments







https://gracewlindsay.com/2018/05/17/deep-convolutional-neural-networks-as-models-of-the-visual-system-qa/

Deep Learning Boom (2012–2018)

Milestones:

AlphaGo (developed by DeepMind) defeated several top human Go players in a series of historic matches:

- AlphaGo v.s. Fan Hui (3-dan professional) in 2015: AlphaGo won 5-0
  Published in Nature (2016), shocked the Go community.

- AlphaGo v.s. Lee Sedol (9-dan professional) in 2016: AlphaGo won 4-1

- AlphaGo v.s. Ke Jie (World #1-ranked player at the time) in 2017: AlphaGo won 3-0

Later Version: AlphaGo Zero

Played only against itself — no human data

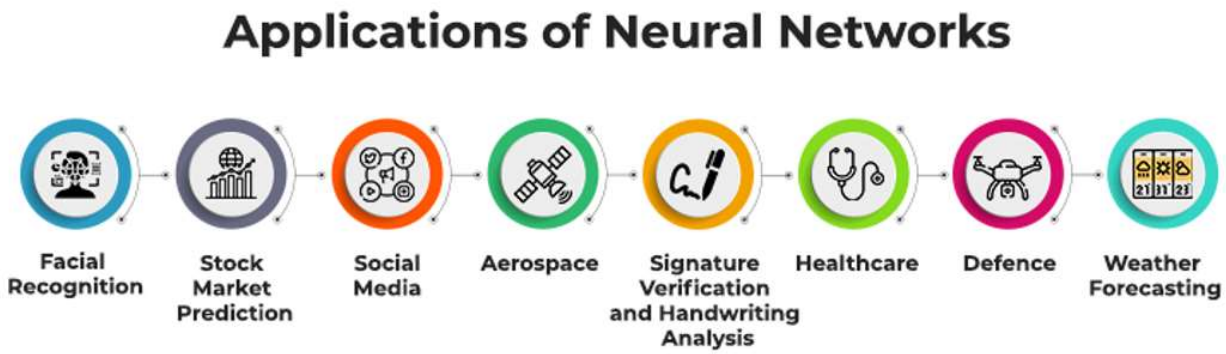
Stronger than all previous AlphaGo versions

Never played against humans (only evaluated in self-play)

Deep Learning Boom (2012–2018)

Impact:

AI entered mainstream industry: vision, speech, translation, medical imaging, etc.



https://www.analyticssteps.com/blogs/8-applications-neural-networks

Generative AI & AGI (2017–Present)

Transformer, a new deep learning framework, was proposed in 2017 by Google Brain, laid the foundation for generative AI.

Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Number of citations: 267,440

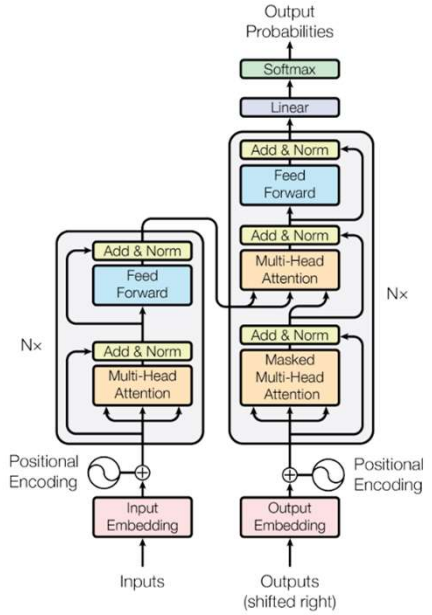


Figure 1: The Transformer - model architecture.

Generative AI & AGI (2018–Present)

Many Large language models (LLMs) were developed based on the transformer framework

◇ OpenAI – GPT Series
Models: GPT-2 (2019), GPT-3 (2020), GPT-4 (2023),
GPT-4o (2024), GPT-5 (2025), GPT-5.1 (2025),
GPT-5.2 (2025) Product: ChatGPT



Anthropic – Claude Series
Models: Claude 1 (2023), Claude 2 (2023), Claude 3 (2024),
Claude 3.5 (2024), Claude 4 (2025), Claude Sonnet 5 (2026),
Claude Opus 5 (2026)



Google DeepMind – Gemini Series
Models: Gemini 1 (2023), Gemini 1.5 (2024), Gemini 2.0 (2024),
Gemini 2.5 (2025), Gemini 3 (2025)



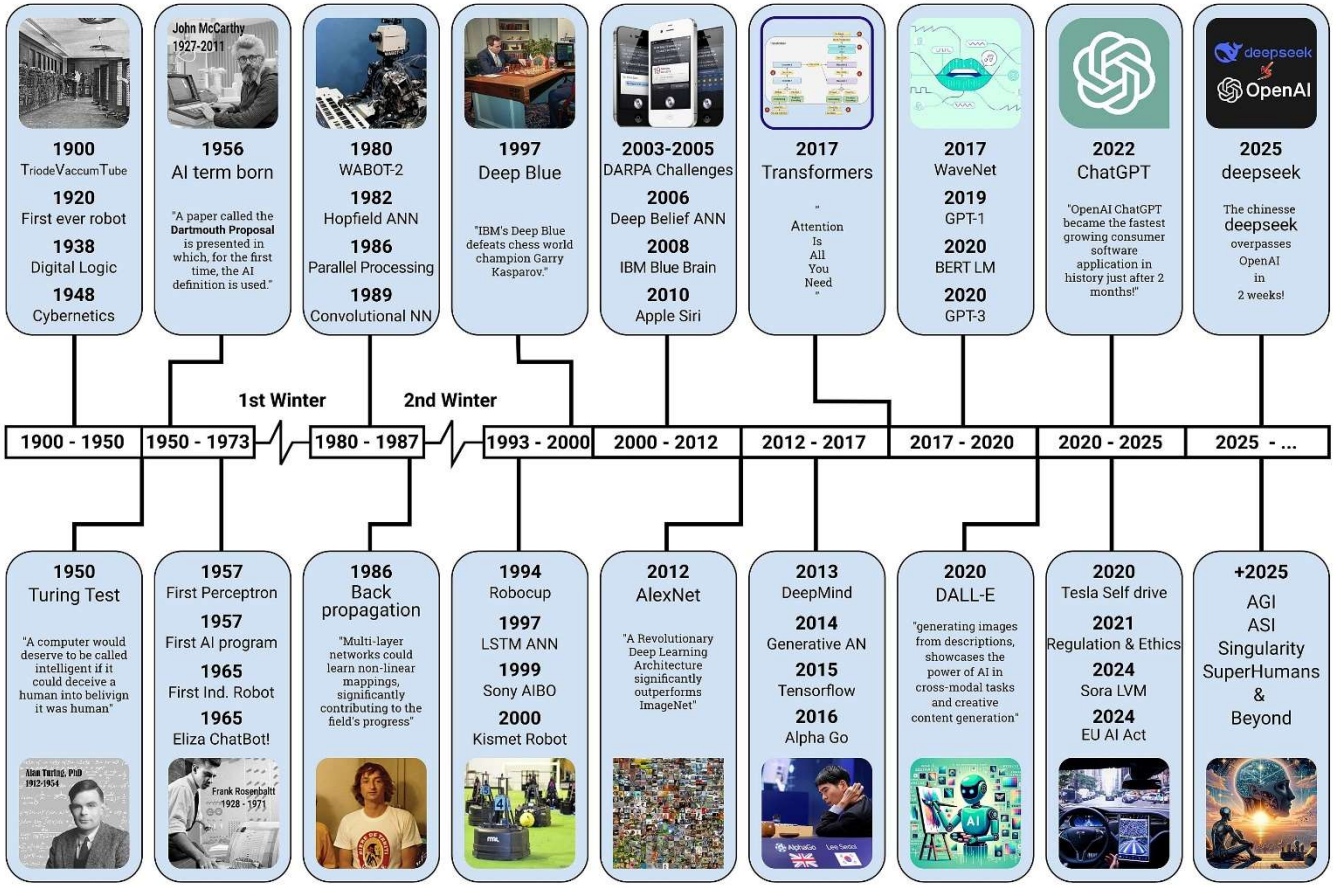
High-Flyer Quant – DeepSeek Series
Models: DeepSeek-V2 (2024), DeepSeek-V3 (2024),
DeepSeek-R1 (2025), DeepSeek-V3.2 (2025),
DeepSeek-V4 (2026) Open-source



Timeline of AI Development

1956–1974	**First Boom**	- Dartmouth Conference - Perceptron (early neural net)	☑ Optimism: Belief that general AI was near ☑ Government funding begins
1974–1980	**First Winter**	- Perceptron limitations (can't solve XOR) - Criticism by Minsky & Papert - Lighthill Report	✗ Technical limits ✗ Failing to meet expectations ✗ Funding cuts
1980–1987	**Second Boom**	- Expert Systems widely adopted in business	☑ Commercial success ☑ Industry investment
1987–1993	**Second Winter**	- Expert systems hard to scale & maintain	✗ Brittleness ✗ Lack of learning ✗ ROI disappoints
1993–2011	**Third Boom**	- Rise of Machine Learning	☑ Statistical methods ☑ More data & computing power ☑ Internet boom
2012–Present	**Fourth Boom**	- **AlexNet** wins ImageNet - Deep learning explosion - LLMs: ChatGPT, Deepseek	☑ Breakthroughs in deep learning ☑ Big data + GPUs ☑ Massive investment & real-world use

Timeline of AI Development



https://en.wikipedia.org/wiki/Timeline_of_artificial_intelligence

PART THREE

Trends & Challenges in AI

Trends & Challenges in AI

Trends:

Multimodal: Cross-modal generation across text, images, audio, and video

Autonomous AI Agents: Evolution from passive chatbots to agents capable of autonomous planning, tool utilization, and complex task execution.

Edge AI & On-Device Models: Efficient Small Language Models (SLMs) running locally on smartphones, PCs, and edge devices.

AI for Science: Deep application of AI in accelerating fundamental scientific discoveries

Trends & Challenges in AI

Challenges:

Hallucinations & Reliability: The risk of models generating plausible but false information, and ensuring accuracy in critical domains.

Security & Misuse: Vulnerabilities like prompt injection attacks, deepfakes, and the spread of automated misinformation.

Explainability & Alignment: Solving the "black box" problem and ensuring AI behaviors align with human values and ethics (mitigating bias).

Environmental Impact: The massive energy consumption and carbon footprint associated with training and running large-scale AI models.

The Future of AI: Opportunities & Risks

Opportunities:

Healthcare, education, automation, science

Risks:

Mass unemployment
AI weaponization
Ethical dilemmas

The development of AI needs to be accompanied by global regulation.



Thank You!